

Raj Khatri

LinkedIn: <https://linkedin.com/in/rajkhatri2295> · **Portfolio:** <https://rajkhatri08.github.io>
GitHub: <https://github.com/rajkhatri08>

Mobile: +91 7252091514
Email: rajkhatrigbpuat@gmail.com

PROFESSIONAL SUMMARY

Final-year Industrial & Production Engineering student building applied GenAI systems. Recently shipped a retrieval-augmented generation system over regulatory documents — vector search, grounded generation, and a governance layer that vetoes expired or unapproved sources — raising retrieval accuracy from 78% to 95% through document-structure chunking. Earlier, at Tata Motors, diagnosed a live Body-in-White production defect by isolating an upstream supplier-side cause through regression-based trend analysis on inspection data. Works in Python, FastAPI, and PostgreSQL.

EDUCATION

B.Tech, Industrial & Production Engineering
College of Technology, G.B. Pant University of Agriculture & Technology, Pantnagar

Uttarakhand, India
August 2023 – August 2027

Senior Secondary (XII), CBSE · Maheshwari Public School, Kota · 75.2%
Secondary (X), CBSE · Peace Public School, Gopeshwar · 97.2%

2022
2020

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL / MySQL), JavaScript, Bash

GenAI & ML: RAG pipeline design, vector search (ChromaDB), sentence embeddings (MiniLM / ONNX), grounded generation & refusal constraints, retrieval evaluation, Scikit-learn

Backend: FastAPI, Pydantic, SQLAlchemy, PostgreSQL, Flask, Next.js, Supabase, REST API design

Data & Tools: Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Excel (Advanced / VBA), Power BI, Git/GitHub, Render, Vercel, Neon

Methods: Root-cause analysis, statistical modeling & hypothesis testing, evaluation-set design, exploratory data analysis, DFMEA, Six Sigma

EXPERIENCE

Tata Motors
Industrial Trainee — Data Analytics

Pantnagar, Uttarakhand **(On-site)**
03 July 2026 – 03 August 2026

Project: "Predictive Quality Intelligence — A Data-Driven Framework for BIW Root-Cause Diagnosis"

- Identified a supplier-side dimensional defect as the root cause of a recurring Body-in-White quality issue, ruling out on-line equipment wear.
- Applied linear regression with R² confidence scoring across ~300 inspection data points, then validated the result by physical measurement on the shop floor.
- Designed a containment procedure deployable without production stoppage or vendor dependency.
- Built an offline, interactive HTML/JS diagnostic dashboard tracking defect trends across 10 monitored quality metrics.
- Authored a 20+ section technical report benchmarking the findings against DFMEA, Six Sigma, and peer-reviewed manufacturing literature.

Voldebug Innovations Pvt. Ltd.
Backend Developer (Part-time)

Remote
22 August 2026 – Present

- Migrating a live K–12 AI-literacy platform from browser-local persistence to a Supabase (PostgreSQL) backend — schema design, row-level security policies, and role-based authentication for student, teacher, and principal accounts.
- Implementing server-side APIs in Next.js (App Router) for XP/badge progression, class-code enrollment, and teacher analytics, replacing client-only state with a shared data layer.
- Establishing database migrations, staging/production environment separation, and a repeatable deployment workflow on Vercel.

Intern

01 July 2026 – 31 July 2026

- Built the platform's first working version — a gamified AI-literacy SPA spanning student, teacher, and principal roles, with a curriculum-linked prompt library and an XP/badge/streak progression system.
- Developed AI-assisted lesson-planning tools and a principal-facing analytics dashboard; implemented Hindi/English bilingual support and downloadable student certificates.
- Authored the phased Supabase + Next.js migration plan now being implemented in the backend role above.

PROJECTS

Source-Governed RAG *Python, FastAPI, ChromaDB, MiniLM/ONNX, PostgreSQL, Next.js · Render / Vercel / Neon*

Live demo

- Built an end-to-end retrieval-augmented generation system over regulatory documents, combining vector search, grounded generation, and a governance layer that vetoes untrustworthy sources.
- Improved retrieval accuracy from 78% to 95% on a labelled evaluation set by replacing fixed-size character chunking with document-structure chunking.
- Architected a governance layer where document approval status and review dates override similarity ranking, preventing answers drawn from expired or unapproved sources.
- Swept every similarity threshold from 0.80 to 1.50 in a calibration script, measured a 73.9% accuracy ceiling, and removed distance-based confidence scoring as insufficiently reliable.

Supply Chain Optimization Engine *Python, Scikit-learn, Pandas, NumPy*

- Trained a Random Forest model on historical demand to generate daily forecasts in place of manual estimation.
- Built replenishment logic around Economic Order Quantity (EOQ) sizing with reorder-point triggers.

CERTIFICATIONS

Forage Job Simulations: J.P. Morgan (Quantitative Research) · Deloitte Australia (Data Analytics) · Mastercard (Cybersecurity)

2026